Lecture 7: Conscious of the Present; Conscious of the Past: Language (cont.); Vision and Memory (full transcript)

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Overview: This lecture finishes the discussion of language by briefly reviewing two additional topics: communication systems in non-human primates and other animals, and the relationship between language and thought. The majority of this lecture is then spent on introducing students to major theories and discoveries in the fields of perception, attention and memory. Topics include why we see certain visual illusions, why we don't always see everything we think we see, and the relationship between different types of memory.

Chapter 1. Non-human Communication [00:00:00]

Professor Paul Bloom: Two follow-ups on yesterday's — I'm sorry, on Monday's lecture. One is that somebody came up after class and asked when the preference for your own language emerges in development and fortunately, [A graduate teaching assistant] studies pretty much exactly this sort of infant understanding. She knew the answer. There's been studies looking at newborn babies finding that pretty much the moment they pop out they favor their own language over other language — over other languages. And this suggests that they are listening while in utero, while in the womb, to the rhythms of their language and developing a preference for it.

A second issue is, I talked very briefly about a court case in which the person was — said at a moment where someone else was pointing a gun at a police officer, "Let him have it!" and a police officer was killed. And that person was charged with murder but I admitted I didn't actually know how things turned out and [a graduate teaching assistant] was kind enough to do extensive research. Well, he went to Wikipedia and [laughter] found out the answer. The answer is he was tried and found guilty for murder. He was then subsequently pardoned. In fact, he was pardoned in 1988, which is really nice except he was executed in 1957. But they did it into a movie. So, it's a movie.

Okay. So, I want to do today, for the first part of the lecture, is continue the language lecture and then move to perception, attention, and memory. And what we had spoken about was — We first talked about universals of language, then moved to some detail about the different aspects of language including phonology, morphology, and syntax. We discussed the ways in which language does the amazing things it does, including the fact that language has used arbitrary science or sounds to convey concepts, and that languages exploit a combinatorial system including recursion to put together these symbols into a virtually limitless set of meaningful sentences. We then talked about development and made some remarks about the developmental time course – talking about the emergence of language from babies to – where babies are really good at learning language to you who are not, whose brains have atrophied, whose language capacities are dead.

Final issue is to shift to animals. Now that we know something about language, we could then ask do animals use — possess the same sort of language? And if not, can they learn it? Now, there is absolutely no doubt at all that nonhuman animals possess communication systems. This has been known forever and is not a matter of controversy. And if you want to use the term "language" to mean

"communication," then the answer is obviously "yes." Dogs and bees and monkeys have language. If you want to use language though in the more technical, narrow sense as anything that has the properties that we discussed earlier, using English and ASL and Spanish and so on as our background, the answer's almost certainly "no."

Animal communication systems fall into sort of one of three categories. Either there is a finite list of calls, so vervet monkeys, for instance, have a small list of calls to convey different warnings like "attack from a snake" or "attack from a leopard." There is a continuous analog signal. So, bee dance, for instance, works on this way. Bee dance communicates the location of food sources but doesn't do it in any syntactically structured way. Rather, the intensity of the dance corresponds to the richness of the food source. And then, you get things like random variation on a theme such as birdsong. But what you don't find in any real sense is phonology, morphology, syntax, combinatorial systems or arbitrary names.

Now, this much is not particularly controversial. There gets to be a lot of controversy though. This is the summary about nonhuman communication systems. It gets more controversial when we get to famous cases of primates trained by humans such as Kanzi, Nim Chimpsky, and other famous primates that you may well have seen on the Discovery channel and other venues. And this is fairly controversial. If you read the Gray textbook, while nothing in it is particularly inaccurate, I think Gray is actually a little bit too credulous, too believing in the claims that have been made about the abilities of the animals. So many scientists argue, for instance, that animals like Kanzi, even if they can be said to be learning words at all, learn very few of them. And it takes them extensive years of training to learn, unlike a normally developing child who could learn a word in a day or a word in an hour. The utterances often have order but this order tends to be very limited and lacks the recursive properties. And in fact, the lack of recursion is not controversial.

Finally, the utterances of chimpanzees — trained chimpanzees are extremely repetitive so what you often see on TV and in documentaries is sort of a sampling. And the sampling could often be very impressive but if you take just what they say at random it tends to look like this. This is typical chimpanzee utterances just taken at random: "Nim eat, Nim eat. Drink, eat, me Nim. Me gum, me gum. Tickle me, Nim play. Me eat, me eat. Me banana, you banana, me you give. Banana me, me eat. Give orange, me give, eat orange, me eat orange." Lila Gleitman once commented that if any normally developing child spoke like this, his parents would rush him screaming to a neurologist.

There's a broader question here, which is, "Why would we ever expect a chimpanzee to learn a human language?" We don't normally expect one species to have the capacities associated with another species. So, bats use echolocation to get around and some birds navigate by the stars, but there's not an active research program seeing if cats can use echolocation or dogs could navigate by the stars. And I think one reason why you might be tempted to think, "well, of course chimps must be able to learn language" is because you might be caught in the grips of some bad ideas about language.

So, one idea is you might say, "Look. Chimps should use language because chimps are so smart." But the response to this is, "they are smart but we know that smart isn't enough." We know that the human capacity for language is not totally a result of smartness. There are smart children who, due to some deficit in their language capacity, don't speak or understand a language. So, the smartness of chimpanzees does not in itself demonstrate that they should be able to learn language.

You might also point out correctly that chimps are our nearest evolutionary relatives, which is right, so you — one would expect on the face of it — it's not unreasonable to expect us to share a lot of abilities with them. On the other hand, we split from them a long time ago and plainly humans are different from chimps. And there was five million years either way and that's more than enough time for a language capacity to evolve.

Now, none of this is to say that the study of nonhuman communication systems isn't interesting. From my own — This is my personal opinion I'll raise here. From my own opinion, the study of the attempts to try to teach chimpanzees, or gibbons, or gorillas, a human language like ASL are misguided. It would be as if a team of monkeys kidnapped a human child and tried to train him how to hoot like a monkey. It might be enjoyable but it does not seem to give us any rich insights. What I think is a lot more interesting is the study of these animal communication systems in the wild. There's a linguistics of human language that has delineated the principles that underlie all human languages. It would be as extraordinarily interesting to attempt the same linguistic program to the other communication systems used in the wild such as the cries of vervet monkeys and bee dance.

Chapter 2. The Relationship between Language and Thought [00:09:16]

So, this brings the section on language to a close but I want to tell you a few things we didn't talk about. One of the problems with an Intro Psych course is we have to whip through a lot of topics very fast. So, if you were to take a course that focused directly on language you might learn, for instance, more about language in the brain, something touched about very briefly in the textbook but something that has a large literature associated with it. Similarly, and related to this, there's language disorders, disorders like aphasias and disorders like specific language impairment and dyslexia. There is the study of language perception and production. How is it that we do this amazing feat of understanding and producing words in a fraction of a second? Where does that ability come from?

There is the study of reading which is, in many ways, different from the study of a language. Remember when Darwin described language as an instinct. He carefully distinguished it from other things that don't come natural to us including reading. And in fact, reading is difficult. Reading is a cultural invention, not every human has it. And unlike language, reading is acquired with tremendous difficulty over many years. On the other hand, reading plainly intersects with language. It's a new way of conveying language, moving out from speech to writing. And the psychology and neuroscience of reading is thus very interesting.

There's bilingualism and multilingualism. The questions people in this room typically are going to be interested in is does it matter for how well you learn language whether you're learning one or two or three or four. How is it that a multilingual encodes all these different languages inside a single brain? And so on. Finally, a very hot issue is that of the relationship between language and thought and I'm actually — A few years ago I taught an entire seminar called "Language and Thought" devoted to precisely this question. And it's a cool question and it could break up into two very general questions. One is, "Is language necessary for abstract thought?" And one way to answer that question is to look at creatures without language like babies and chimpanzees and see how smart they are. It might be that they're not — that they're very smart, in which case it would suggest you don't need language for

abstract thought. On the other hand, it might be that they have certain cognitive limitations, which would suggest that language is essential for abstract thought.

Then there's the related question. Even once you know a language, does the structural properties of the language that you know affect the way you think? And the claim that the language you know affects how you think is sometimes described as linguistic relativity or the Sapir-Whorf hypothesis. So for instance, there's a lot of research looking at speakers of different languages such as English versus Korean and seeing whether structural differences in these languages affect how you think. Now, some of this work is discussed in the readings, the book — the Gray textbook, and the selections from The Norton Anthology. And this makes up — again, I've showed this to you on Monday — your reading response where you have to address this question and take your best shot at answering it.

Chapter 3. Question and Answer on Language [00:12:39]

What are your questions about language? Yes.

Student: [inaudible]

Professor Paul Bloom: The question was raised, "Some people learn languages easier than others and how do we explain this?" And the answer is you could ask the question both with regard to first language learning – so some children learn language very quickly, some are very slow – and also with regard to second language learning. Some of you are breezing through your second language requirement here at Yale. Others are struggling and miserable. And there's considerable variation. There's the story of Einstein who was very slow to learn language and didn't speak at all until he was four. And in fact, he was a — He said his first words when all of a sudden he was having supper with his parents and he put down the spoon and he said, "The soup is too hot." And his parents stared in astonishment and said, "You've never spoken before." And he said, "Well, up to now everything's been fine." [laughter] It's not a true story. [laughter]

The question of why and where these differences come from, nobody really knows and it's surprisingly hard. There's a slight advantage for being female. Girls are slightly more advanced in language than boys but it's not a big one and you need a hundred people to just see it statistically. There's a big genetic factor. If your parents learned language quickly and learned other languages quickly, you are more likely to. But an understanding of the brain bases of these differences or the cognitive bases or the social bases is just — is largely an open question. Yes.

Student: What happens when parents [inaudible]

Professor Paul Bloom: This is actually more the norm around the world than the situation in the United States where kids are exposed to a single language. What happens is children learn both languages. Children are very good, as adults are, of distinguishing different languages on the basis of their sound system and their rhythms so they don't typically confuse them. And then they just learn more than one language. And that's actually more the average state of affairs around the world. Yes.

Student: You said that people who are right-handed learn languages [inaudible]

Professor Paul Bloom: The question is about the hemispheric specialization for language. And I don't have actually much more to say than what I said before, which I agree is deeply unsatisfying. If you're

right-handed, language is probably in the left side of your brain. How many people here are left-handed? For you we don't know. It varies. Some of you have it in the left side. Some of you have it in the right side. For some of you it's kind of diffuse. Now, why is this? And in fact, why are some people right-handed and others left-handed in the first place? Those are really good questions. Yes.

Student: [inaudible]

Professor Paul Bloom: Yes. I'll — Yes, that's — I'll answer that question. And unfortunately, it's going to be the last one and then I'll go to vision. The question is, "Does learning more than one language cause you to learn them slower than just learning one language?" And it would stand to reason that it would. There's a finite amount of mental resources. If I'm just learning English, I use all of it for English. And if I'm learning English and Spanish I kind of got to split. And you'd expect them to be each learned slower. It's one of the surprises of the study of language development that that common-sense view does not appear to be true. Children learning more than one language seem to show no deficit relative — in each of their languages, relative to a child learning just one language. In other words, if I am just learning English and I'm a kid and you're learning English and Spanish and you're a kid, you'll reach the milestones in English the same time I will. Your extra learning of Spanish doesn't seem to affect you. There doesn't seem to be any detriment for learning multiple languages.

Another question which comes up is, "Is there any cognitive deficit?" In other words, some people have argued that learning multiple languages sometimes harms children in certain ways. This is a claim that's been made in Quebec, for instance, over the debate over how children should be taught English and French. It does not appear to be the case. There appears to be, as far as we know, no down side to learning many languages when you're young. Does that answer your question?

Chapter 4. Introduction to the Complexity of Perception and Expectation [00:18:06]

I want to move now to the topic that will take us through today and through the beginning of next week – perception, attention, and memory. And I'm putting them together instead of treating them as separate lectures because there's a sense in which they're the same story. You see a scene. You see this scene and you're looking at it and you're perceiving it. It's coming through your eyes and you're interpreting it and you see something. You see a man and you see a house. If you were to shut your eyes, you could still hold that scene in memory. And a week later, if I'm to ask you about that, "What season was it?" you would do pretty well. This is the story I want to talk about – how we do this.

And in the course of this I want to make a series of claims that go something like this. For perception, I want to first persuade you the problem of perception's hard and that successful perception involves educated and unconscious guesses about the world. For attention, I want to suggest that we attend to some things and not others and we miss a surprising amount of what happens in the world. For memory, there are many types of memory. The key to memory is organization and understanding. And you can't trust some of your memories.

How many of you remember where you were at 9/11? Many of you are wrong. And I am never going to persuade you of this because you have certain memories. And you could tell the story. Everybody could tell the story where they were when the towers went down. But clever psychologists on September 12

said, "Let's do a study." And they asked people, "Where were you yesterday when you heard the news?" And they told them. And then they went back to them later, a year later, two years later, and said, "Tell me about what happened September 11." And they said, "I remember totally where I was. I have a very — " And then — And often the story was wrong. There is a lot like that which we're going to talk about. And the biggest moral then — so, I put it really, in really big print — We are often wrong about our experiences, both of the present and of right now. So, let's start with perception.

There is a story — I went to graduate school at MIT and there was a story there about Marvin Minsky who is the A.I. [artificial intelligence] guru. He — If you've heard the words — the phrase "artificial intelligence," that was him. And if you heard the claim that people are nothing more than machines made of meat — also him. Well, there's a story where he was doing work on robotics and he was interested in building a robot that could do all sorts of cool things that's like a robot. And the story goes the robot had to among — had to write — had to see the world. It had to be able to pick up things and recognize people and see chairs and navigate its way and Minsky said, "That's a tough problem. It's going to take a graduate student a whole summer to figure it out." And he assigned it to a graduate student for a summer project.

Visual psychologists, perception psychologists, love that story because the study of computer vision and robotics vision and the attempts to make machines that can identify and recognize objects has been a profound failure. There is, at this point, no machine on earth that could recognize people and objects and things at the level of a really dumb one-year-old. And the reason why is that it's a much harder problem than anybody could have expected. Well, what makes it such a hard problem?

Well, one reason why you might think it's an easy problem is you say, "Okay. We have to figure out the problem of how people see. Well, here's what we do." [pointing to a slide that caricatures the inside of a person's head as containing a little man, the real "you," sitting in a control room watching a television monitor that is connected to the larger-head's eyeballs] You're in — You're over there and here's your eye. And somehow it has to get to this television monitor and then you look at it and that'll solve the problem of how you see. So, sometimes people say, "Hey. I hear the eye flips things upside down. I guess this guy [the guy in your head] is going to have to get used to looking at things upside down. That's an interesting problem." No. That's not the way to look at it because that doesn't answer any questions. That just pushes the question back. Fine. How does "he" see? We're not answering anything.

Similarly, although the Terminator's [the cyborg assassin from the movie "The Terminator"] view of the world may correspond to that [showing a slide of what vision looks like to the Terminator – a series of gauges and numbers], that doesn't solve any problem of how he actually sees. So, he has all these numbers shooting out there. Well, he has to read the numbers. He has to see this. This [pointing to some icons at the edge of his slide] is my iTunes. [laughter] That's inadvertent.

Here's the right way to think about perception. You got the eye, which is very ugly and bloody, and then around here you have the retina. And the retina is a bunch of nerve cells. And the nerve cells fire at — for some stimulus and not others. And from this array of firings, "firing... not firing... firing... not firing," you have to figure out what the world is. So, a better view is like this. The firings of the neurons could be viewed as an array of numbers. You have to figure out how to get from the numbers to objects and people, and to actions and events. And that's the problem. It's made particularly a difficult problem because the retina is a two-dimensional surface and you have to infer a 3D world from a two-

dimensional surface. And this is, from a mathematical point of view, impossible. And what this means is that there — For any two-dimensional image there is an indefinite number of three-dimensional images that correspond to it.

So for instance, suppose you have this on your retina, an array of light shaped like that [referring to a slide portraying a square and an irregular polygon that could be a square that is tilted backwards in space]. What does that correspond to in the world? Well, it could correspond to a thing just like that that you're looking for or it could correspond to a square that's tilted backwards. And so, you have to figure out which is which. And the way we solve this problem is that we have unconscious assumptions about how the world works. Our minds contain certain assumptions about how things should be that enable us to make educated guesses from the two-dimensional array on to the three-dimensional world.

And I purposefully did not make the slides available for this class ahead of time because I don't want people to cheat, but there are several points where you could look at the slides and confirm that some of the things I'm going to tell you are actually true. And I want to give you three examples. One is color. And I'm going to conflate here color and brightness. The other is objects. The other is depth.

First, the problem of color. How do you tell a lump of coal from a snowball? Well, that's a lump of coal and that's a snowball, and it's from Google images. How do you know which is which? Well, a lump of coal you say is black and a snowball is white. How do you know? Well, maybe you have on your retina — Your retina responds to sort of color that hits it. It's oversimplified, but let's assume that this is true. So, this is black coming out and that's white and that's how you tell. But in fact, that can't be right. It can't be right because objects' color is not merely a matter of what material they're made of but of the amount of light that hits it. So, as I walk across the stage I fall into shadow and light, and none of you screams out, "Professor Bloom is changing colors!" Rather, you automatically factor out the change in illumination as this is happening.

And this could actually be quite striking. So, you see this display over here. Take a look at those two blocks. [a slide portrays two blocks of different luminance, one under a table, one in the middle of a lit room.] I take it you see this one [the object under the table] as lighter than that one. You do. You might imagine this is because this strip [the block under the table] is lighter than this [the block out in the open] but it isn't. They're the same. And you won't believe me until you actually print it out and take a look, but they are in fact the same. I'll show it to you. And you could say I'm tricking you but this is the way it works. There's the close-up. So, remember we're comparing this and this [the two blocks]. Now, let's take away other parts of the environment and you'll see they're the same. [As Professor Bloom covers the background surrounding both of the blocks they suddenly appear to be the same color as one another.]

Now you say, "But hold it. This can't be the same as this" but the answer is — goes like this. We know shadows make surfaces darker. We don't know this like "Here's something I know." Rather, we know this in that it's wired up in our brains. So when we see a surface in shadow we automatically assume that it's lighter than it looks, and we see it as lighter. And you could show this by removing the cues to the shadow. And you see it as it really is. And this is an illustration of how the information to your eyes is just one bit of information; the degree of light coming from a single source is one bit of information that you use to calculate certain assumptions and come to a conclusion.

Here's a different kind of example: Objects. You see this [a picture of a man walking down a path, in front of his house] and you automatically and intuitively segment it into different objects. You segment it into a man and a house and birds and trees. How do you do this? It turns out, to program a computer to segment a scene into different objects is hugely difficult and the question of how we do it is, to some extent, unknown. But one answer to this question is there are certain cues in the environment that are signals that you're dealing with different objects. And these cues are often described as Gestalt principles.

So, one example is "proximity." When you see things that are close to each other, you're more likely than not to assume that they belong to the same thing. There's "similarity." That display [a group of many objects that are all the same shape, but all the objects on one side have a different texture than those on the other side] could correspond to an indefinite number of objects but you naturally tend to see it as two. You do one with one texture pattern, the other with the other texture pattern. "Closure." The fact that this is a closed square here suggests it's a single object [referring to a line drawing of a square overlapping a circle]. "Good continuation." If you had to judge, this [referring to a picture of two overlapping lines, line AB and line CD] could just as well be two shapes, one that runs from A to C, the other one that runs from D to B. But you don't tend to see it that way. Rather, you tend to see it as one that runs from A to B, the other one that runs from C to D. "Common movement." If things move together they're a single object. And "good form." You see the object over there [two overlapping and perpendicular rectangles]. In the absence of any other information, you might be tempted to say that's a single thing, a plus sign maybe. This [pointing to two overlapping but non-perpendicular rectangles], because it has lousy form, you're more tempted to say it's two things, one thing lying on top of each other.

And these are the sort of cues, expectations; none of them are right. There's cases where they could all fool you. But these are useful cues that guide our parceling of the world, our segmenting of the world into distinct objects. Here they are summarized [pointing to a slide showing all the cues on the same page]. And here's a case where they fool you [pointing to a slide showing a Kanizsa Triangle, an illusory triangle induced by three incomplete circles]. So you might think, if you're suggestible, that there is a triangle here. And this is a case where there are certain cues driving you to think that there's a triangle here. There is, however, no triangle here. If you cover up these little Pacmen here, the triangle goes away. Similarly, there is no square in the middle [referring to a picture of a Kanizsa Square]. There is no square. It's very Matrix. And these are illusions because these are cues that there should be a square there, the regularity of form.

Finally, "depth." You see this [the picture of the man walking away from his house] and you don't — You see it on one level as a flat thing. Another level you look inside the picture and you see, for instance, the man is in front of the house. You look at me and you see the podium. And if you have a terrible neurological disorder you see this strange creature that's half podium leading on to a chest and up to a head that's sort of — the top of him is wiggling and the podium staying still. If you are neurologically normal, you see a man walking back and forth behind a podium. How do you do that? Well, this is really a problem because, I could give you a technical reason why vision is hard, but crudely, you got a two-dimensional retina and you have to figure out a three-dimensional world. How do you do it? And the answer once again is assumptions or cues. There are certain assumptions the

visual system makes that aren't always right and in fact, in cases of visual illusions, can be wrong but will guide you to perceive the world in a correct and accurate way.

So for example, there is binocular disparity. This is actually a sort of interesting one. This is the only depth cue that involves two eyes. If I look at you [a student sitting in the front row] pretty close, the image I get here [pointing to his right eye] and the image I get here [pointing to his left eye] are somewhat different while — or I have to focus my eyes together to get the same image. If I look at you in back, they're almost identical because the further away, given the two eyes that are static, the closer the images look. And it's not, again, that you say to yourself, "Oh. Back there an orange. It's the same image in my right eye and my left eye. You must be far away." Rather, unconsciously and automatically you make estimations on how far people are in depth based on binocular disparity.

There is "interposition." How do you know I'm in front of the podium and the podium's not in front of me? No. How do you know the podium's in front of me? Well, from where I'm standing it's right. How do you know the podium is in front of me? Well, because I'm walking here and then it cuts into me. And unless I'm going through a grotesque metamorphosis, what's happening is it makes sense to say I'm moving behind the podium. Interposition. You take the guy. How do you know the guy is standing in front of the house? Well, because there is — you see all of him and he's blocking a lot of the house.

There's relative size. How far away am I? Well, if you looked at me and you had to estimate how far away I am, part of the way you'll figure that out is you know how tall a human's supposed to be. If you thought that I was fifty feet tall, you would assume I'm further away than I am. And so, your judgments on size dictate your judgments about distance. Usually, this cue isn't necessary but if you look at the Empire State Building — If you go into a field and you see a tower and you look, your judgment of how far away the tower's going to be depends on your knowledge of how tall a tower should be. If it's this tall, you say, "Oh. It must be — " And then you'd be surprised. There's texture gradient, which I'll explain in a second, and linear perspective, which I'll also explain in a second.

Texture gradient goes like this. Remember the problem we had before. How do you know if that thing [a spotted rectangle that's tilted backwards] is this object [a spotted rectangle standing upright] or an object in and of itself? Well, the answer is things with textures will show themselves because the textures will get smaller from a distance. Now, logically, this could still be a single thing standing upright with just dots going up smaller. But the natural assumption is the reason why the dots recede in this regular fashion is because it's receding in depth.

Classic illusion – the Mueller-Lyer illusion. People will see this as longer than this [referring to one of two arrow-like lines, one with both ends pointing inward, the other with both ends pointing outward]. It's not. If you don't believe me, print it out and measure it. Related to the Ponzo illusion, once again people see this one [showing a picture of two gradually converging lines crossed by several horizontal lines, like a train-track receding in the distance] as — you get illusions named after you when you discover these — this one [a horizontal line at the top] as longer than this [a horizontal line at the bottom]. Again, it's not.

What's going on here? Well, the top line looks longer even though it isn't. And one explanation for why is, these other lines in the scene cause your visual system to make guesses about distance. And then you correct for distance by making assumptions about size. If you have two lines — You'll get — We'll

get in more detail in a second, but if you have two lines and they take up the same amount of space on your retina, but you believe that one is 100 feet away and the other's 50 feet away, the one that's 100 feet away you will see as bigger because your brain will say, "Well, if it takes up just this much space but it's further away, it must be bigger than something that's closer and takes up that much space." And that's what goes on here.

For the top line, for the Mueller-Lyer illusion, we assume that this is further away and this is closer based on the cues to distance. And the cue is factored in. And because we assume that this is further away, we assume it must be bigger to take up the same space as this which is closer. Similarly for the Ponzo illusion. There's linear perspective. Parallel lines tend to recede in distance. If this top one is further away than this but they take up the same size in your eye, this one must be bigger and you see it as bigger. And the book offers more details on how these illusions work.

I'm going to end with an illusion that I'm not even going to bother explaining. I'll just show it to you because you should be able to, based on thinking about these other illusions, figure it out. It was developed by Roger Shepard. Well, you know that. And they are called Shepard tables [pointing to a picture of what looks like two simple dining tables, or desks. One that looks longer and skinnier than the other]. And the thing about it is, these look like two tables. If you ask people — You don't frame in terms of here's a lecture on visual perception. You ask people, "Which of these tables would be easier to get through a door if you have a thin door?" People would say the one on the left. This one looks sort of thicker and harder to get through. This one looks longer and leaner. In fact, they're the same size. What I mean by that is that this [rectangle] is exactly the same as this [rectangle].

Now, I'm going to prove it to you by showing you something which took me — on the computer which took me about seven hours to do. And nobody's going to believe it because I could have faked it. But if you want, print it out and do it yourself. You just take a piece of paper, put it on here. Then you move it [he demonstrates that a piece of paper, cut to be the same size as one of the tables, fits perfectly over the other table] and [they're] the same. I showed it to somebody and they called me a liar. Anyway, you could do it yourself in the privacy of your own home or study. But what I'd really like you to do after you do it is say, "Okay. Fine. Why does this one look longer and thinner than this one?" And the answer is the same answer that will explain the Mueller-Lyer illusion and the Ponzo illusion, having to do with cues to depth and the way your mind corrects the perception of depth. And that's all I have to say at this point about perception.

Chapter 5. Linking Attention and Memory [00:39:15]

I want to move now to attention and memory and I'm going to treat attention and memory together. We are fascinated with memory and, in particular, it's particularly interesting when memory goes wrong. It's particularly fascinating what happens in cases of amnesia. So for example, I need a volunteer who is willing to do a little bit of acting, a very little bit, an incredibly little bit. [a student volunteers] Excellent. Okay. So well, you just stay there. So pretend you have amnesia. Okay? What's your name?

Student: I don't know.

Professor Paul Bloom: Perfect. I'm really glad you said that. That's the wrong answer because you don't have total amnesia. You still remember English. Okay. It's very clever. Okay. So you couldn't have lost all your memories. You have English. You [pointing to a different student] — So we'll do you. What's your name? Oh. He looks puzzled but he still maintains bowel and bladder control so he hasn't forgotten everything. [laughter] Now, I always lose the third volunteer in that demo.

So, what I'm saying is that memory is a hugely broad concept. It includes autobiographical memory, which is what we standardly think. That's a perfectly rational response. When I say somebody's losing their memory, "Oh. I have a movie about somebody losing their memory," you don't imagine a person in diapers. You imagine the person walking around, having sex with cool people and saying, "Where am I?" And [laughter] so what you imagine is you imagine them losing their autobiographical memory, their sense of self. But of course, knowing English is part of your memory and knowing how to stand and knowing how to chew and swallow are all things that you've learned, that you've — that have been molded by experience.

There's another distinction which is going to come in regarding amnesia, which is there's broadly two types of amnesia. They often run together, but one type of amnesia is you lose your memory of the past. Another type of amnesia — That's the Matt Damon amnesia. Another type of amnesia though is you lose the ability to form new memories. And here's a film of a man who had exactly this problem. [film playing]

He was a world-renowned choir director and he suffered viral encephalitis which led to brain damage which destroyed most of his temporal lobes, his hippocampus, and a lot of his left frontal lobe. It could be — It could have been worse in that he retains the ability to talk. He seems to be — He's not intellectually impaired. He just can't form new memories and so he lives in this perpetual "now" where just nothing affects him and he feels — This has not always happened. There's more than one of these cases and it doesn't always happen like this, but he feels continually reborn at every moment. And we'll return to this and then ask what's going on here. But there's a few themes here.

I want to, before getting into detail about memory, I want to review some basic distinctions in memory when we talk about memory. So crudely, you could make a distinction between sensory memory, short-term memory, which is also known as working memory, and long-term memory. Sensory memory is a residue in your senses. There's a flash of lightning. You might see an afterimage. That afterimage is your sensory memory. There's somewhat of a longer echoic memory for sounds. So as somebody is talking to you even if you're not paying attention you'll store a few seconds of what they're saying, which is sometimes, when somebody's talking to you and you're not listening to them and they say, "You're not listening to me." And you say, "No. You were talking about — " and pick up the last couple of seconds from echoic memory. There's short-term memory.

Anybody remember what I just said? If you did, that's short-term memory — spans for a few minutes. And then there's long-term memory. Anybody know who Elvis is? Do you know your name? Do you know where you live? Your long-term memory store that you walk around with and you're not going to lose right away. When we think about amnesia in the movie sense, we think of a certain loss of long-term memory associated with autobiographical personal events.

There is a distinction between implicit and explicit, which we'll talk about it in more detail. But explicit, crudely, is what you have conscious access to. So, what you had for dinner last night. You could think back and say, "I had this for dinner last night." Implicit is more unconscious. What the word — what certain word — what the word "had" means, how to walk, how to ride a bicycle, that you might not be able to articulate and might not even be conscious of but still have access to.

There's a distinction between semantic memory and episodic memory. Semantic memory is basically facts, what a word means, what's the capital of Canada, and so on. Episodic is autobiography, is what happened to you. That Yale is in New Haven is semantic. That you went on vacation away from New Haven last week, it would be episodic. There is encoding stores and retrieval, which refers to different levels of what happens in memory. Encoding is getting the memory in, as when you study for a test or you have an experience. And storage is holding the memory. And retrieval is getting the memory out.

Finally, retrieval is often broken, conveniently, into recall versus recognition, where recall is when you just pull it out of memory and recognition is when you recognize what corresponds to something in the past. Anybody remember what color tie I had on two days ago? Oh. Okay. Well, that would be impossible to remember but if I asked you, "Is it purple or is it orange?" that would be much easier. [laughter] Now, you could break up, crudely, the memory into stages. So you start that sensory memory is just the stuff that comes in leading to short-term memory, leading to long-term memory. And this stage theory is something which we'll discuss in more detail. But this leads us to the issue of attention.

How do you get memory from your sensations, from what you're hearing? I'm speaking to you. You're hearing me. How does it ever get in to the other systems? What decides what's remembered and what's not? There's all sorts of things happening to you now. The seat of your chair is pressing against your butt. You wouldn't say, "Oh. I want to remember this forever. The seat's pressing against my butt." [laughter] Your neighbor is exuding a certain sort of smell. You're thinking about something. Your eyes follow him. Not everything gets in memory. You'd go mad if you tried to remember everything. You can't. So, what determines what gets into memory? Well, one answer is "attention does."

And attention is — could be crudely viewed as a flashlight, a spotlight on experience that willingly zooms in on something and makes it memorable. Attention has certain properties. Some things come from attention — to attention effortlessly and automatically. Here's an example. You're going to see an array of letters here. One of them's going to be green. When you see the green one, please clap. [laughter] No, not this green one. [laughter] There's going to be another slide. Okay. You're ready now. [students quickly find the one green "x" amongst a background of black "x"s. Okay. Now find — Not that "o" [laughter] but there's going to be an "o." When you see it clap. [students quickly find the "o" amongst a background of "x"s] Okay. Sometimes it's work. Find the red "o." [students are much slower at finding the red "o" amongst a background of black "o"s and red "x"s] [laughter] It's harder.

Sometimes attention is involuntary. I need a volunteer. And all I want to do is I want to show you colors on the screen and I'd like you to name the colors as they come out. [pointing at a student] Do you want this?

Student 1: I'm colorblind.

Professor Paul Bloom: Oh. [laughter] The first one is easy. See. This is — You have to just go down the colors [on the slide are a series of different colored rectangles. The student must name the colors]. Anybody? Okay.

Student 2: Red, green, blue, black, green, blue, red, blue, black, red.

Professor Paul Bloom: Excellent. [applause] Okay. Now these. These will be words but just name — Okay, you. Just name the colors. [this slide contains color words that are written in the same color that they spell. Again, the student must name the color of the font]

Student 3: Green, red, blue, black, blue, red, green, black, red, blue.

Professor Paul Bloom: Perfect. Now, we'll go back to you, same deal, words. [now the slide contains a list of color words that are written in a different colored font then what they spell. The student must name the font color.]

Student 4: Red, blue, green —

Professor Paul Bloom: No, no, no. Huh uh. Don't — I know you can read. The colors.

Student 4: Okay. Sorry. Okay. Blue, green, red, green, black, green, blue, black, red, [laughter] blue, black, red, black. [the student struggles to name the color of the font without accidentally saying the name that is written]

Professor Paul Bloom: Very good actually. [laughter] That's known as the Stroop effect. Being an expert reader, as you are, your knowledge of reading, your attention to what the words meant, subverted your desire to do the task. You couldn't make that go away even if you wanted to. If somebody gave you $1,000 to read this as fast as you read this, and as fast as you read this, you'd be unable to. You can't block it.

There is some work — There are some interesting discoveries about attention. I have a demonstration here. I'd like people actually — It's important — Some of you may have seen this before. It's important for you to be silent throughout it. What you're going to see is you're going to see two teams of basketball players. One of them is going to have white T-shirts. The other one will have black T-shirts. They'll be passing balls back and forth. What I'd like you to do is count in your head how many passes the white team does with the ball. [the video shows several people passing a basketball back and forth while, at one point, a person in a gorilla suit walks across the scene] [laughter] What number did people get? Okay. Did anybody notice anything unusual? [laughter] Did anybody not notice anything unusual? Okay. Some people did not notice anything unusual. Those who didn't see anything unusual, watch this again and just watch it. [laughter] About 50% of people when counting, who have never seen this before don't notice anything. But then when you're not counting it's kind of obvious what you're missing. [laughter] And this is one demonstration among many of the fact that when you're attending to something you have a very small window of attention and you lose the focus on other things.

Here's another different example. I'd like people to watch a movie and pay attention very closely to what happens in the movie and try to remember this. [The movie shows a conversation between two people. Each time the camera cuts from one angle to the next something about the scene changes.] How many of you noticed something odd in that movie? How many of you didn't? Okay. Now,

everybody look at the scarf, the color of the plates and the food, among other things. [the same movie plays again] The phenomena, in general, has been called "change blindness." And what it is is we tend to be — when there's a focus of attention focused in a certain way, we tend to be oblivious to other things that go on in the environment. Often it is, in fact, quite difficult when there's a change in scene to notice what changes and what stays the same.

So, in this final demo, there's just going to be two pictures flicking. Could you clap when you see what's different between the two pictures? [applause] [laughter] I myself am terrible at these and so I have a lot of sympathy. How many people never saw it? [laughter] Good. That's very impressive. [laughter] One more time with a different one. [applause] Did anybody not see it? Be honest. I'll give you another try. [applause] Okay. I'll put you out of your misery. [laughter] This is work by Dan Simons and it's part of an extraordinarily interesting body of work on what's known as "change blindness." And what this means is, the phenomena is, we have a very narrow focus of attention and huge changes can happen that we are oblivious to. This is why, in movies, there are so many — so much difficulty with continuity changes.

Dan Simons is also famous for having brought this outside of the laboratory in some classic experiments and I'm trying to get the film corresponding to them. What he did was that he did this great study in the Cornell campus where he was — where what happened is they would get some unsuspecting person walking through campus and some guy would come over and say, "Excuse me, Sir. I'm lost. Could you help me with directions?" And have a map and then the person would say, "Sure." And then there'd be two construction workers holding a door. And these guys were going to rudely bump between these two characters and then the experimenter gets switched with another guy. So now, when these two guys walk away, the subject is standing there with an entirely different person. [laughter] What's interesting is nobody notices. [laughter] They notice if the person changes sexes. "Didn't you used to be a woman?" [laughter] And they notice if the experimenter changes races, but most other changes they're oblivious to.

There's another experiment. I think Brian Scholl did this one but it may have been Dan Simons where what happens is a subject comes in to the lab. They say, "If you're going to do an experiment with us, you need to sign the human subject form." Hands him the form, the experimenter. The subject signs the form. The experimenter takes the form and says, "Thank you. I'll put it down here." Goes down here and then a different person pops up. [laughter] People don't notice. And there's a certain level on which we're oblivious to changes. What's weird is we don't see — we don't think we are. We think we see the world as it is and we don't know — notice that when we're attending to something; everything else gets blanked out.

And so about 50% of people who have never seen this demo before, the gorilla demo, they don't notice the gorilla. And there's — you couldn't imagine anything more obvious. The gorilla study was actually done a very long time ago. And it was originally done in a different way but I'll show it to you just because this is the original study and now that you all know what to expect — Oh, not that one. Oops. [demo playing] Nope. [demo playing] That's actually — If you looked at that quickly, it's a current Yale professor. Oh. I'm never going to get my DVD back. Anyway, I'll show you the other demo on — next week. I will. [laughter]

Chapter 6. Question and Answer on Attention and Memory [00:58:04]

Any questions about attention and memory at this point? Yeah.

Student: [inaudible]

Professor Paul Bloom: Yeah. Why does it work that way? Why is it — Why do things that become very practiced become automatic and involuntary? It's a good question. I don't know. We know that they do. We know that once you — that you can't not read once you know how to read. You also can't not listen. If I'm talking to you and I'm extremely boring, but I'm talking to you, it's very hard not to listen. You can't shut off your ears. You could put your fingers in them but you can't shut off your — You also can't shut your eyes without actually shutting them. You can't say, "This is a disgusting movie. I'm not going to attend to it." [laughter] So, that's not answering your question. It's just saying that your observation is a right one and a more general one. When you're good at something and you're over-practiced, it becomes involuntary and you cannot stop it. Okay. Well — Oh. One more in back. Yes.

Student: [inaudible]

Professor Paul Bloom: What now? Sorry. Over there. Yeah.

Student: [inaudible] right before his accident?

Professor Paul Bloom: Did he remember things before his accident? Yes. He had some amnesia of events before his accident but he did remember things. He knew his name and he knew other things about his life. Okay. I'll see you next week.